

Generation Interconnection System Impact Study Report

For Queue Project Q18-0955

Meadow Lake Wind Farm Llc 100 kV

November 19, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0955. The Interconnection Customer proposes a 188.5 MW wind facility interconnecting at the Meadow Lake Wind Farm Llc 100 kV substation in the Illinois area. The request was received on March 02, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 311 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,949,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Weed Jct - Meadow Lake Wind Farm Llc 100.0 kV line
Contingency: Beach Haven - Meadow Lake Wind Farm Llc 100.0 kV
Loading: 139.5% (Rating: 124 MVA)
DFAX: 100.0%
Pre-project loading: 13.0%
2. Facility: Weed Jct - East F 100.0 kV line
Contingency: Beach Haven - Meadow Lake Wind Farm Llc 100.0 kV L
Loading: 137.6% (Rating: 118 MVA)
DFAX: 100.0%
Pre-project loading: 21.8%
3. Facility: Beach Haven - Meadow Lake Wind Farm Llc 100.0 kV
Contingency: Meadow Lake Wind Farm Llc - Weed ct 100.0 kV
Loading: 132.6% (Rating: 130 MVA)
DFAX: 100.0%
Pre-project loading: 12.4%
4. Facility: Beach Haven - Bloomingdale 100.0 kV
Contingency: Meadow Lake Wind Farm Llc - Weed Junction 100.0 kV
Loading: 118.4% (Rating: 134 MVA)
DFAX: 100.0%
Pre-project loading: 22.3%
5. Facility: Norwich Substation - Barrett 500.0 kV line
Contingency: Trego - Beaver 500.0 kV
Loading: 101.6% (Rating: 3,374 MVA)
DFAX: 13.8%
Pre-project loading: 100.8%
6. Facility: Barrett - Wolfe 500.0 kV
Contingency: Trego - Beaver 500.0 kV
Loading: 100.9% (Rating: 3,201 MVA)

DFAX: 9.8%
Pre-project loading: 100.3%

7. Facility: Norwich 230/1kV Transformer Unit 1
Contingency: Norwich - Barrrt 500.0 kV L
Loading: 100.7% (Rating: 995 MVA)
DFAX: 7.0%
Pre-project loading: 99.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Meadow Lake Wind Farm Llc - Weed Jct 100.0 kV	Rebuild line with higher-capacity conductor	\$5,851,000	100% (\$5,851,000)
Weed Jct - East F 100.0 kV line	Rebuild line with higher-capacity conductor	\$1,995,000	100% (\$1,995,000)
Beach Haven - Meadow Lake Wind Farm Llc 100.0 kV	Rebuild line with higher-capacity conductor	\$2,056,000	100% (\$2,056,000)
Beach Haven - Bloomingdale 100.0 kV	Rebuild line with higher-capacity conductor	\$4,800,000	100% (\$4,800,000)
Barrett - Norwich Substation 500.0 kV Line	Reconductor line	\$23,426,000	1% (\$248,000)
Barrett - Wolfe 500.0 kV Line	Reconductor line	\$88,783,000	1% (\$923,000)
Norwich 230/1 kV Auto	Replace transformer with higher-rated unit	\$7,036,000	1% (\$76,000)
	Total allocated to project		\$15,949,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.